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Team 6
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Individual Process

This week, my primary contribution for System Demo 3 was implementing and demonstrating a complete inverse kinematics (IK) engine for the hexapod robot. I built a Python-based IK solver from the ground up, including the single-leg IK solver, the full hexapod body model with coordinate frame transforms, and three interactive visualization/demo modes: single-leg servo control, full 6-leg tripod walk, and joystick controller simulation. This represents the core software that will drive all leg motion on the physical robot.

Inverse Kinematics Solver

The IK solver computes the three joint angles (coxa, femur, tibia) needed to place a leg's foot at a desired (x, y, z) position in the leg-local coordinate frame. The leg-local frame has its origin at the coxa joint, with $+X$ pointing outward from the body, $+Y$ pointing left, and $+Z$ pointing up.

The solver works in three steps:

1. **Coxa angle** (q_1): Computed via `atan2`(y, x) to rotate the leg horizontally toward the target.
2. **Projection to vertical plane**: The horizontal distance from the coxa to the target is reduced by the coxa length, giving the femur-to-foot distance $r = \sqrt{d_h^2 + d_v^2}$ in the leg's vertical plane.
3. **Femur and tibia angles** (q_2, q_3): Solved using the law of cosines on the femur–tibia–foot triangle, with reachability checking (the target must satisfy $|L_f - L_t| \leq r \leq L_f + L_t$).

The solver also includes graceful degradation: if the target is slightly out of reach, the distance r is clamped to the reachable range so the leg extends as far as possible without crashing the solver. This is important for real-time animation where small numerical overshoots can occur during gait transitions.

Using the finalized leg dimensions from Dillon's CAD model — coxa: 254 mm (10 in), femur: 228.6 mm (9 in), tibia: 594.4 mm (23.4 in) — the solver produces valid joint angles for the full workspace of each leg.

Servo Angle Mapping

A critical piece of the pipeline is converting IK joint angles (in degrees, relative to the leg's neutral configuration) into servo command angles that can be sent to the Spektrum S6510 servos via the PCA9685 PWM driver. The mapping accounts for:

- **Per-leg direction flags**: Right-side legs have their coxa servos mirrored (-1 direction) because the servos are mounted in opposite orientation.
- **Per-joint calibration offsets**: Adjustable offsets for fine-tuning each servo's zero position during physical calibration.
- **Safe clamping**: All servo outputs are clamped to 30°–150° (the 120° mechanical travel range of the S6510 with 1000–2000 μs PWM).

The reverse mapping (`servo_to_ik_angles`) is also implemented, enabling the interactive demo to convert slider positions back to IK angles for visualization.

Hexapod Body Model

The `Hexapod` class models the full 6-leg robot with body-frame coordinate transforms. Each leg is mounted at a specific position and angle on the body (three legs per side, evenly spaced along the 812.8 mm body length). The key functionality includes:

- **Body-to-leg transform**: Converts foot target positions from the body coordinate frame to each leg's local frame using a 2D rotation by the leg's mounting angle. This is the critical

transform that allows us to specify walking motion in body coordinates while the IK solver operates in leg-local coordinates.

- **Batch IK solve**: Solves IK for all 6 legs simultaneously and returns a flat list of 18 servo angles.
- **UART command generation**: Formats the 18 servo angles into the communication protocol designed in ILR03: `M:a0,a1,...,a17\n`.

Demo Mode 1: Single Leg with Servo Sliders

The single-leg demo (Figure 1) provides an interactive 3D visualization of one leg with three servo sliders (coxa, femur, tibia). The user can manually adjust each servo angle and see the leg move in real time. A "Play" button animates a forward stepping motion that shows the full stance/swing cycle:

- **Stance phase** (first half): The foot stays on the ground and moves backward smoothly using a cosine profile, simulating the leg pushing the body forward.
- **Swing phase** (second half): The foot lifts off the ground with a sinusoidal arc and swings forward to the starting position.

The orange dashed line in Figure 1 shows the complete foot trajectory, and the red vertical markers indicate the stride range limits. During animation, the servo sliders sync to the computed values, so the user can observe how the servo angles change throughout the gait cycle.

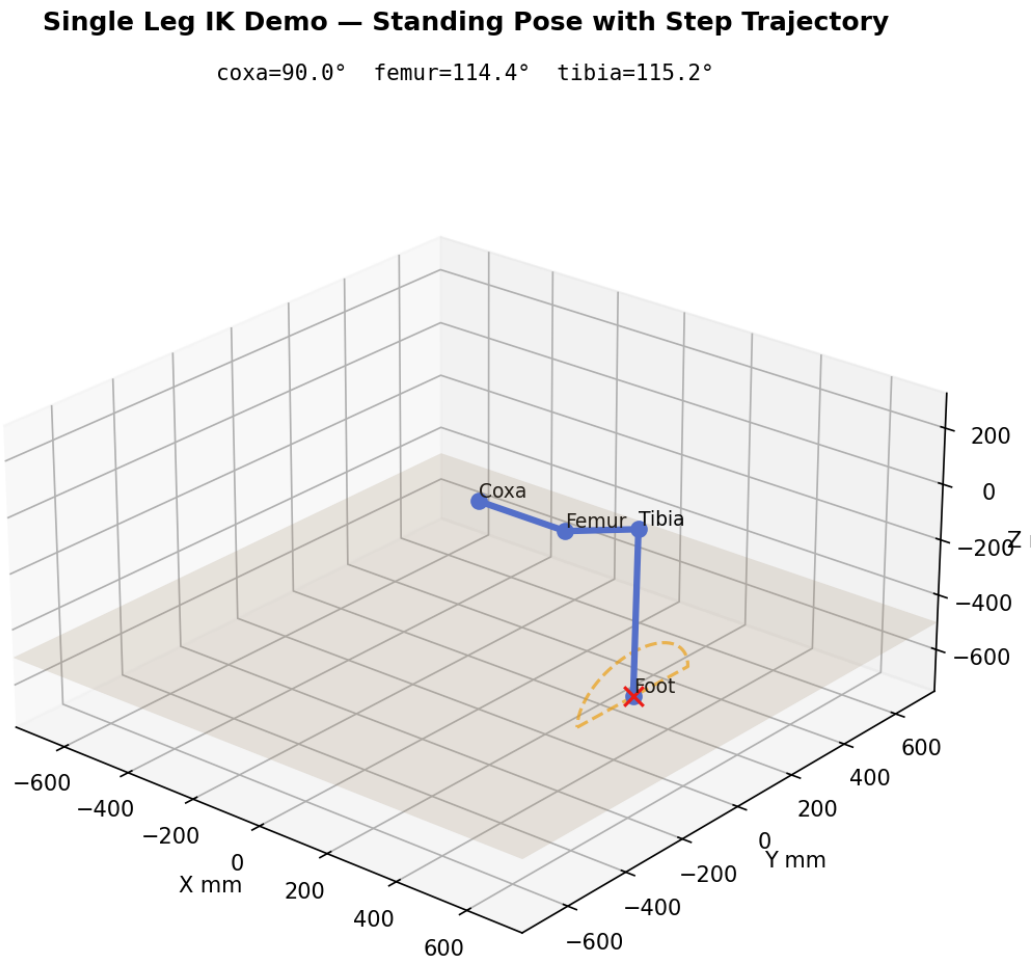

Figure 1: Single-leg IK demo showing the standing pose with the step trajectory (orange dashed line). Joint labels (Coxa, Femur, Tibia, Foot) and servo angles are displayed. The ground plane is shown in tan.

Demo Mode 2: Full 6-Leg Tripod Walk

The full hexapod demo (Figure 2) animates all 6 legs performing a tripod gait — the most common and stable walking pattern for hexapod robots. The legs are divided into two groups:

- **Group A** (legs 0, 2, 4 — L1, L3, R2): Move in phase together.
- **Group B** (legs 1, 3, 5 — L2, R1, R3): Move 180° out of phase with Group A.

At any point during the walk, three legs are in stance (on the ground, pushing backward) while the other three are in swing (lifted, moving forward). This ensures the robot always has a stable tripod of support. The stride length is computed to be collision-safe: $stride = (leg_spacing - 40\,mm)/2 \approx 61\,mm$, with a 40 mm safety gap between adjacent legs' workspaces.

The demo displays real-time servo angles for all 18 servos (3 per leg × 6 legs) in the upper-left corner, directly matching the UART command format. Each leg is rendered in a distinct color for easy identification.

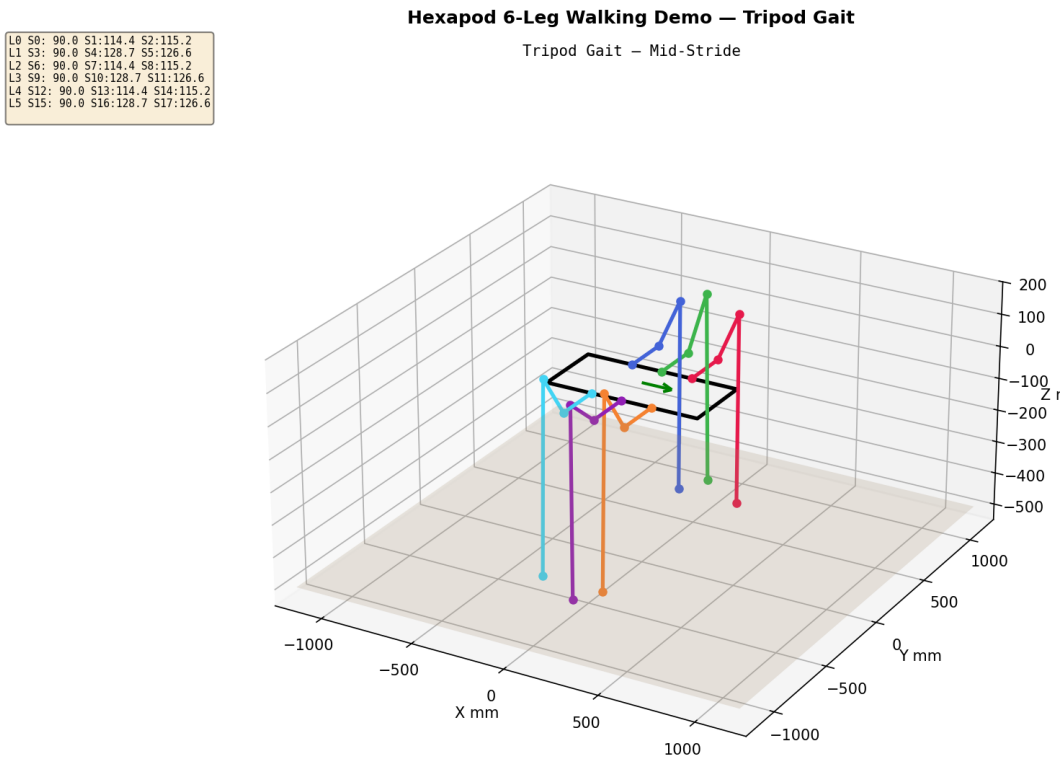

Figure 2: Full hexapod 6-leg walking demo with tripod gait at mid-stride. The body outline (black rectangle) and heading arrow (green) are shown at center. All 18 servo angles are displayed in the upper left. Each leg is color-coded: L1 (red), L2 (green), L3 (blue), R1 (orange), R2 (purple), R3 (cyan).

Demo Mode 3: Joystick Controller

The controller demo (Figure 3) simulates joystick input with two sliders: Forward/Backward and Turn Left/Right. The hexapod responds in real time to the combined input:

- Pure forward/backward input produces a straight walking gait.
- Pure turn input produces a turning-in-place motion where left-side and right-side legs stride in opposite directions.
- Combined input produces curved walking paths.

The display shows the current mode (STANDING, FORWARD, BACKWARD, TURN LEFT, TURN RIGHT) and the live UART move command string, demonstrating the full pipeline from user input through IK to servo commands. This mode directly validates the controller integration path: the same command format will be sent from the Raspberry Pi to the ESP32 over UART.

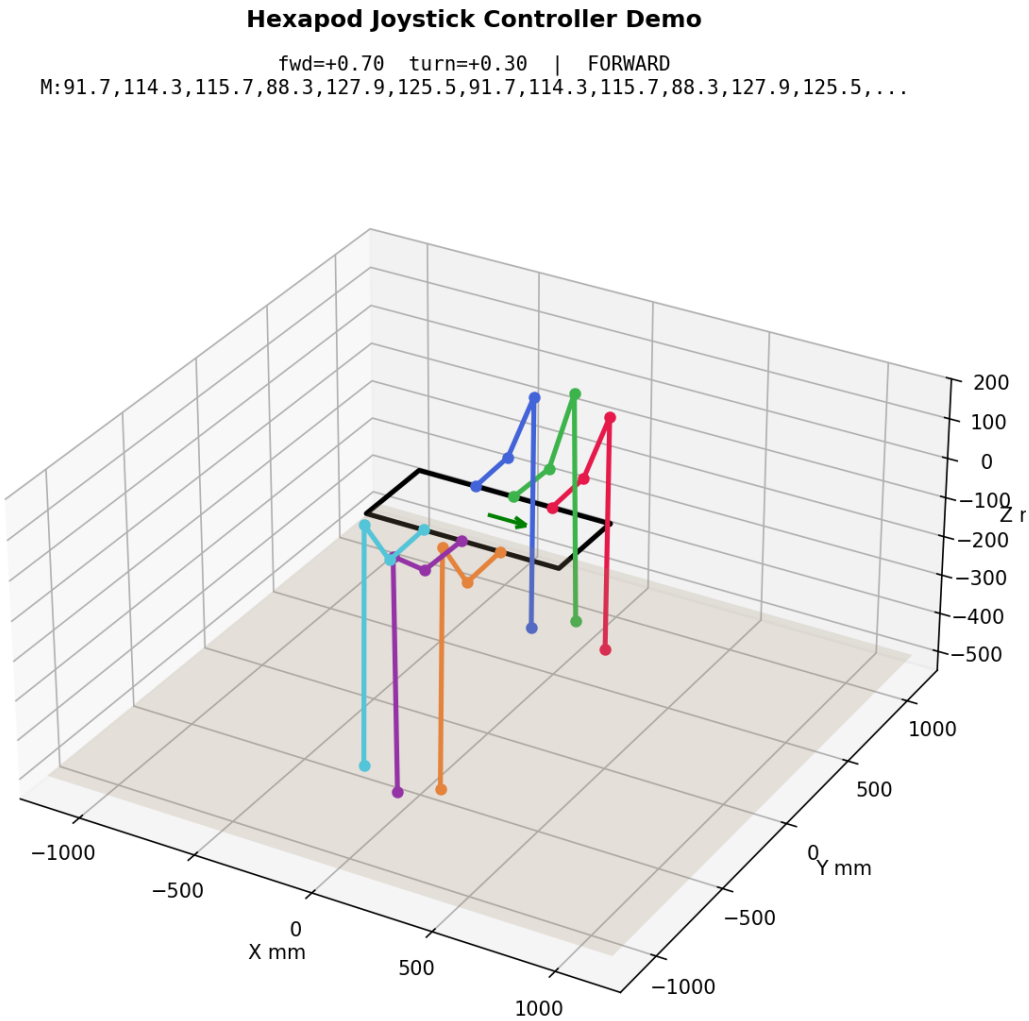

Figure 3: Joystick controller demo showing the hexapod moving forward with a slight right turn. The UART move command is displayed at the top, showing the 18 servo angle values that would be sent to the ESP32.

Challenges

The primary challenge was getting the coordinate frame transforms correct between the body frame and each leg's local frame. Each leg is mounted at a different position and angle on the body, and the IK solver expects inputs in the leg-local frame where $+X$ is always "outward." Initially, I had the body-to-leg rotation inverted for right-side legs, which caused those legs to mirror their motion instead of walking in the correct direction. Debugging this required careful visualization of each leg's local axes and verifying that the coxa angle produced the expected horizontal rotation for each mounting configuration.

Another challenge was tuning the gait parameters to avoid leg collisions. With six large legs (total reach $\approx 1077\,\text{mm}$ each) on a relatively compact body ($812.8\,\text{mm} \times 381\,\text{mm}$), adjacent legs

can easily interfere during the swing phase. I computed the minimum safe stride based on the leg spacing along the body and added a 40 mm safety margin. The lift height is set to 80% of the stride length, which provides sufficient ground clearance during the swing phase.

A third challenge was the tibia angle convention. The IK solver naturally produces q_3 as the angle between the femur and tibia (where π = fully folded, 0 = fully extended), but the servo expects an angle where 90° = straight leg. The conversion required careful sign handling and a 90° offset, and I had to verify it with forward kinematics to ensure the visualization matched the expected physical behavior.

Team Work

The team made strong progress this week across all subsystems. Each member delivered a concrete, demonstrable result for System Demo 3:

- **Mithun:** Built a physical static model of a single leg using the laser-cut birch plywood parts (Figure 4). This prototype validates the mechanical design — the leg dimensions, joint clearances, and servo mounting positions all match the CAD model. The physical leg will be used for testing servo-driven motion once the electronics are wired.
- **Thomas (me):** Implemented the complete IK engine and three interactive demo modes as described above, demonstrating single-leg IK with servo control, full 6-leg tripod walking, and joystick controller simulation.
- **Leena:** Connected user-input devices (XR controller) to the ESP32 controller over Bluetooth. This establishes the wireless control link from the operator to the robot's embedded processor.
- **Dillon:** Finalized the CAD model to approximately 95% completion in SolidWorks, including the body frame, all six leg assemblies, and the seated platform. The remaining 5% covers minor mounting bracket adjustments and cable routing features.

Figure 4: Physical static model of a single hexapod leg built by Mithun. The leg is constructed from laser-cut birch plywood with servos mounted at each joint (coxa, femur, tibia). The three segments match the designed dimensions: coxa (254 mm), femur (228.6 mm), and tibia (594.4 mm).

Together, these deliverables demonstrate end-to-end progress: we have a validated physical leg prototype, a working IK engine that generates correct servo commands, a wireless controller link, and a nearly complete CAD model ready for fabrication.

Plans

Looking ahead, with System Demo 6 targeting significant wiring clean-up and a final power solution, and System Demo 7 targeting a dress rehearsal, our near-term priorities are focused on integration and physical assembly. For the next milestone, I plan to:

- Deploy the IK engine onto the Raspberry Pi and test it driving real servos on the physical

leg prototype via the ESP32 UART link, validating the full software-to-hardware pipeline.

- Integrate Leena's Bluetooth controller input with the IK engine's controller mode, enabling live joystick-driven motion on the physical robot.
- Work with Mithun to calibrate the per-servo offsets and direction flags using the physical leg, updating the `config.py` parameters based on measured servo behavior.
- Begin testing multi-leg coordination on the physical chassis once Dillon's final CAD parts are fabricated and assembled, starting with a 2-leg then 6-leg walking test.
- Assist with wiring and power system integration as purchased components arrive, working toward the wiring clean-up target for System Demo 6.